

ONE LABEL, THREE NUMBERS

each reads as “time to depletion”

arithmetic — not a forecast
reserves ÷ production
reserve-life ratio

an index — not a stock ratio
a fitted trend vs its own minimum
trend index

a pressure scale — not depletion
log margin ÷ mortality
removals-only time

public data: G3P · USGS · RAM

THE TYPED LEDGER

each moiety in its own currency

biomass → **funds** → **biodiversity**

typed fluxes — never summed
from the incidence structure
conservation

outflows vanish with their donor
positivity

readouts, not conserved mass
services

turnover · frozen rate · hitting time
three depletion times, not one

NO WEIGHT HIDES A DEFICIT

the vector carries the certificate

for every weight ≥ 0
weighted aggregate ≥ 0



one floor fails
componentwise

no phantom mass
double counting

each claim carries its own predicate
an aggregate ≥ 0 is not each floor ≥ 0
no weighting certifies all floors

Typed Flux Ledgers and Depletion Arithmetic: Conservation, Componentwise Diagnostics, and the Semantics of Depletion Horizons